

I

EXPERIMENTS ON QUANTIZATION

1. Introduction

A prominent characteristic of present-day physics is that many of the quantities used to describe physical phenomena are quantized. That is, such quantities cannot take any one of a continuum of values, but are restricted to a set (perhaps an infinite set) of discrete values. Common examples are the intensity of radiation of the electromagnetic field, the energy of atomic systems, and the electric charge. Strong evidence for such quantization is obtained from the three experiments that will be described in this chapter:

(a) Millikan's experiment by which the charge on individual oil droplets is measured. The experiment shows that the charge is always an integer multiple of the smallest charge observed; this is identified with the charge of the electron.

(b) The Frank-Hertz experiment on the excitation by electron bombardment of atomic vapors. It is found that only for discrete bombarding energies is such excitation possible, and the first excited state of the mercury (Hg) atom is thus measured.

(c) The photoelectric effect, in which the emission of electrons from a cesium (Cs) surface under the influence of electromagnetic radiation in the visible spectrum is studied. It is then seen that in this process each individual electron absorbs from the radiation field (which has a frequency ν) energy $h\nu$, where h is Planck's constant.

2. The Millikan Oil Drop Experiment

2.1 GENERAL

In 1909, R. Millikan reported a reliable method for measuring ionic charges. It consists of observing the motion of small oil droplets under the influence of an electric field. Usually the drops acquire a few electron charges and thus conventional fields impart to them velocities that permit isolation of a drop and continuous observation for a considerable length of time; further, the mass of the oil droplet remains almost constant (there is very slight evaporation) during these long observation times.

In principle, if we measure the force due to the electric field E ,

$$F_e = qE = neE \quad (2.1)$$

we can obtain ne ; repeating this measurement for several (or the same) drops but with different values of the integer n , we can extract the charge of the electron e .

The electric force can be measured either by a null method—that is, by balancing the drop against the gravitational force—or as will be described here, by observing the motion of the drop under the influence of both forces. Oil droplets in air, acted on by a constant force, soon reach a terminal velocity given by Stokes' law,[†]

$$F = 6\pi\eta av \quad (2.2)$$

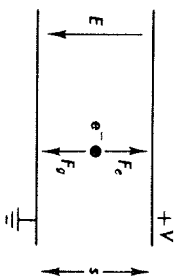
where a is the radius of the (assumed spherical) droplet, η the viscosity of the oil, and v the terminal velocity. To obtain the radius of the drop (needed in Eq. 2.2) we observe the free fall of the drop; the gravitational force is

$$F_g = \frac{4}{3}\pi a^3(\rho - \sigma)g \quad (2.3)$$

with ρ and σ the density of air and oil and g the acceleration of gravity.

Schematically, as shown in Fig. 1.1, the apparatus consists of two parallel plates which can be alternatively charged to a constant potential $+V$, $-V$, or 0. The drop is then observed (with a telescope) and the time t it takes to travel through a distance d is measured. Let F_+ denote the force on a negatively charged drop with electric field up (time t_+ ; electric force aiding gravity) and F_- the force with electric field down (time t_- ; electric

Fig. 1.1 Forces on a charged oil drop between the plates of the Millikan apparatus.



force opposing gravity). Then

$$F_{\pm} = \pm ne(V/s) - \frac{4}{3}\pi a^3(\rho - \sigma)g = 6\pi\eta ad(1/t_{\pm}^{(n)})$$

$$F_0 = -\frac{4}{3}\pi a^3(\rho - \sigma)g = 6\pi\eta ad(t_0)^{-1} \quad (2.4)$$

where the sign conventions hold if t is considered >0 when the drop moves up, $t < 0$ when it is moving down. A convenient method of analysis is to write Eq. 2.4 as

$$\frac{1}{t_{\pm}^{(n)}} = \pm An - B \quad A = \frac{Ve}{6\pi\eta ad}$$

$$\frac{1}{t_0} = -B \quad B = \frac{2}{9} \frac{a^2(\rho - \sigma)g}{\eta d} \quad (2.5)$$

so that A and B can be easily determined.[†]

Indeed, a plot of $1/t_{\pm}^{(n)}$ against n reveals the linear relationship and the fact that only integer values of n appear, proving that the drop has acquired one, two, three, or more electric charges of value e , and never a fraction of that value. Thus we have clear evidence that the ionic charge picked up by the oil drops is *quantized*. Furthermore, the absolute value of this minimal electric charge is in good agreement with inferred measurements of the charge carried by the atomic electrons,[‡] and therefore is accepted as the most accurate value of the charge of the electron.

2.2 THE EXPERIMENT

The apparatus used in this laboratory (Fig. 1.2) consists of two parallel brass plates $\frac{1}{4}$ in. thick and approximately 2 in. in diameter, placed in a lucite cylinder held apart by three ceramic spacers 4.7 mm long. This assembly is in turn enclosed in a cylindrical brass housing with provisions for electrical connections and containing two windows, one for illumination of the drops and one for observation. The top plate has a small hole in its

[†] These expressions are in cgs units so that V must be expressed in statvolts if e is to be obtained in esu.

[‡] As in e/m experiments, shot noise measurements, etc.

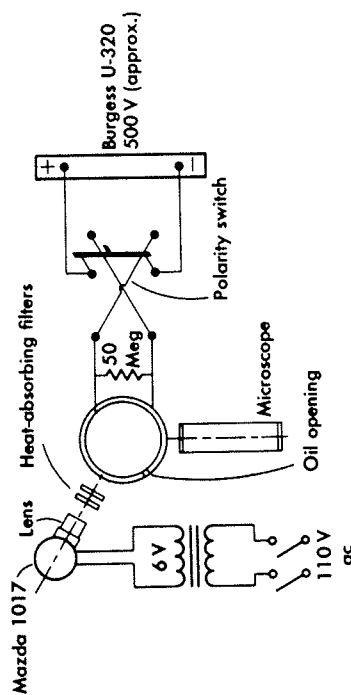


FIG. 1.2 Millikan oil drop experiment schematic of the apparatus.

center for the admission of the oil drops, which are produced by spraying oil with a regular atomizer.

To charge the plates, a 500-V battery and a reversing switch are used; the plates are shunted by a 50-megohm resistor to prevent them from remaining charged when the switch is open. For observations a 10-cm focal length microscope is used (Cenco 72925), while illumination is provided by a Mazda 1017-watt lamp and condensing lens. To avoid convection currents inside the apparatus, a heat-absorbing filter (Corning infrared-absorbing) is placed in the illuminating beam.

The plates should be made perpendicular to the gravitational field by means of the three leveling screws at the base of the apparatus and a level placed on the top plate. Being a cosine error, the deviation introduced by an angular displacement of the gravitational component from perpendicular by 8 degrees is 1 percent. A value for the plate spacing s may be obtained by using the stage micrometer. The micrometer should be focused on a wire inserted in the oil hole in the center of the top plate, and the crosshair of the micrometer should be moved along the length of the wire. Several measurements should be taken and their results averaged.

The velocities are determined by measuring with a stop watch the time required for the droplet to cover a specified number of divisions of the microscope scale. Care must be taken to avoid drafts and vibrations in the vicinity of the apparatus; for that reason and because of Brownian motion, the drop may wander or be displaced out of the field of the microscope. It may then be necessary to *reposition* the microscope between measurements on a single drop. Moreover, the drop should be kept in focus to avoid parallax errors.

Both the microscope and the light source may be adjusted by viewing a small wire inserted in the oil hole. The light should be adjusted so that

2. The Millikan Oil Drop Experiment

the focal point is somewhat ahead or behind the wire and the wire is more or less evenly illuminated. To light the scale, a small light is placed next to the slit just ahead of the eyepiece of the microscope. The actual distance to which a scale division corresponds may be found by using a microscope slide on which a subdivided millimeter scale has been scratched.[†] The eyepiece focus of the microscope should not be changed during a run, since moving the eyepiece changes the effective distance of the scale. (To bring the drop back into focus the entire microscope should be moved.)

It is important to be *sparing* in the amount of oil sprayed into the chamber. In addition to gumming up the interior more quickly, large quantities create so many particles in the microscope field that without excessive eyestrain it is virtually impossible to single out and follow a single droplet.

Under the influence of gravity, droplets will fall at various limiting speeds. If the plates are charged, some of the drops will move down more rapidly, whereas others will reverse their direction of motion since in the process of spraying some drops become positively charged and others negatively charged. By concentrating on one drop which can be controlled by the field, and manipulating the sign of the electric field so that this particular drop is retained, it is possible to remove all other drops from the field. The limiting velocity is reached very quickly and the measurement should be started near the top or bottom of the plate. Measurement should be completed before the drop has reached a point in its travel where application of the reverse potential is insufficient to save the drop from being "gobbled up."

The density in air of the oil used was $\dagger 0.883 \pm 0.003 \text{ gm/cm}^3$. It is desirable to take measurements in the shortest possible time since, as previously mentioned, the mass of the drop changes through evaporation.

It is obviously desirable to make measurements on as many different charges on the same or different drops as possible. Thus after four or five measurements of $t_+^{(n)}$, $t_-^{(n)}$, and t_0 have been taken, the charge on the drop must be changed; this is accomplished by bringing close to one of the windows a radioactive source (10 to 100 μCi of Co^{60} will do).[§] The droplet should be brought close to the top plate and allowed to fall with the *field off*; on its way down it will sweep up a few ions created by the source. This can be checked by occasionally turning the field on to see if the charge has changed; rarely will a drop pick up any charge when the field is on.

The battery voltage should be checked with a 1-percent resistor divider and potentiometer; microscope calibration should be checked before and

[†] Note that the focal length of the microscope must not be changed, but instead the slide should be brought into the focal plane.

[‡] This may be found by a simple measurement.

[§] $\text{Ci} \equiv \text{curie}$.

after the measurements; the same holds true for air temperature and pressure, which are needed for a correction to Stokes' law.

Indeed, when the diameter of the drop is comparable to the mean free path in air, the viscosity η in Eq. (2.2) should be replaced by

$$\eta(T) = \eta_0(T) \left[1 + \frac{b}{aP} \right]^{-1} \tag{2.6}$$

where $\eta_0(T)$ is the viscosity of air as a function of T (Fig. 1.3), $b = 6.17 \times 10^{-4}$, P is the air pressure in centimeters of mercury, and a is the radius of the drop in centimeters (of the order of 10^{-4} cm). In analyzing the data it is convenient to calculate a_0 by letting $\eta = \eta_0(T)$ in Eq. 2.5; a_0 is then inserted in Eq. 2.6 to obtain $\eta(T)$ and thus a more accurate value for a .

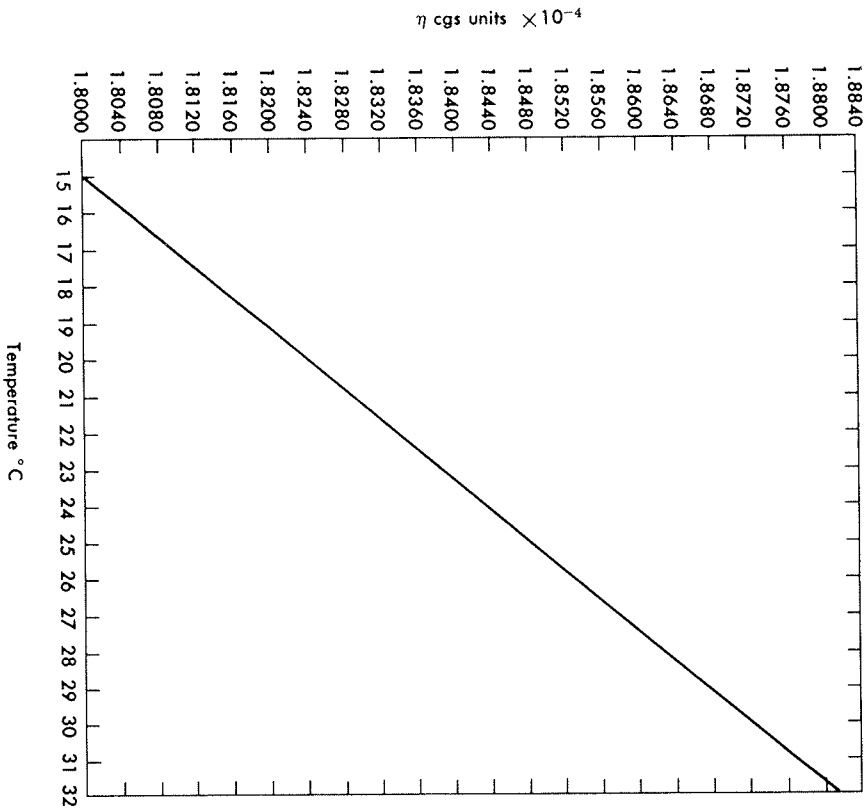


Fig. 1.3 Viscosity of dry air as a function of temperature.

TABLE 1.1
DATA ON MILLIKAN OIL DROP EXPERIMENT

$\bar{t}_0$	$\bar{t}_-^{(n)}$	$\bar{t}_+^{(n)}$	$(1/t_-)-(1/t_0)$	$(1/t_+)-(1/t_0)$	n	$\frac{(1/t_+)-(1/t_-)}{2n}$
Drop 1						
-27.9	+8.69	-5.65	-0.142	+0.150	1	-0.146
-29.6	+1.36	-1.18	-0.813	+0.770	5	-0.158
-28.2	+3.66	-3.00	-0.299	+0.308	2	-0.152
-29.3	+0.75	-0.716	-1.362	+1.368	9	-0.152
-29.4	+2.35	-1.97	-0.473	+0.460	3	-0.155
$1/t_0 = -0.0346$ $a = 4.91 \times 10^{-6}$ cm $\eta = 1.60 \times 10^{-4}$						
Drop 2						
-24.22	+3.98	-3.071	-0.285	+0.291	2	-0.144
-25.75	+9.73	-5.65	-0.137	+0.143	1	-0.140
-25.4	+2.5	-2.12	-0.432	+0.440	3	-0.145
-25.22	+9.67	-5.42	-0.145	+0.143	1	-0.144
-25.22	+4.1	-3.07	-0.286	+0.288	2	-0.143
-24.4	+1.73	-1.73	-0.538	+0.618	4	-0.144
-24.4	+9.95	-6.02	-0.126	+0.141	1	-0.133
$1/t_0 = -0.0400$ $a = 5.06 \times 10^{-6}$ cm $\eta = 1.59 \times 10^{-4}$						
0.1433						

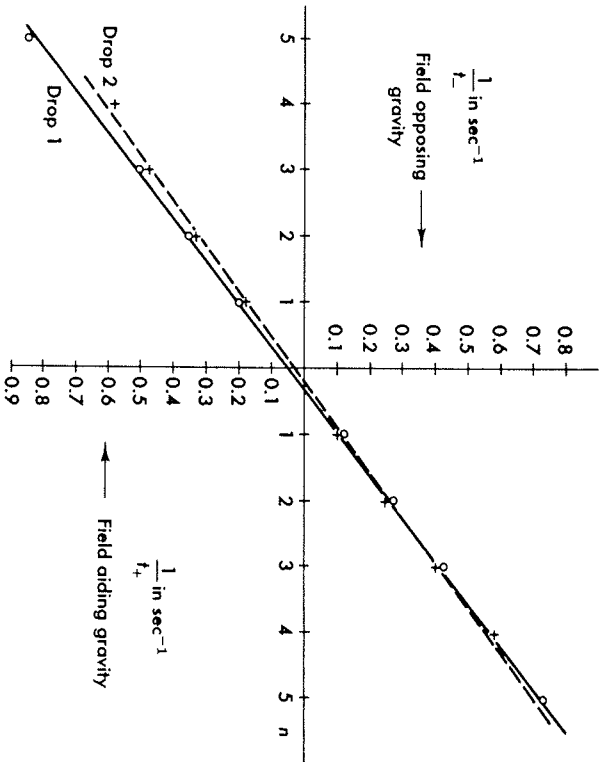


Fig. 1.4 Plot of $1/t_+$ and $1/t_-$ versus n where n is an integer.

2.3 ANALYSIS OF THE DATA

Table 1.1 is a sample of data obtained by a student.† Two drops were used and several charges were measured; for each charge six measurements were performed and averaged as shown in Table 1.1.

The pertinent parameters for these data were

Distance of fall	$d = 7.63 \times 10^{-2}$ cm
Temperature	$T = 25^\circ \text{C}$
Pressure	$P = 76.01$ cm Hg
Density	$\rho' = (\rho - \sigma) = 8.82 \times 10^{-1}$ gm/cm ³
Potential	$V = 500$ V = 1.666... statvolts
Plate separation	$s = 4.71 \times 10^{-1}$ cm

A plot of the data and the linear least squares fit are shown in Fig. 1.4. From the least squares fit we obtain (see Eq. 2.5)

$$A_1 = 0.1533 \pm 0.0011 \quad B_1 = -0.0380 \pm 0.0053$$

$$e = (4.90 \pm 0.1) \times 10^{-10} \text{ esu}$$

$$A_2 = 0.1439 \pm 0.0017 \quad B_2 = -0.0402 \pm 0.0038$$

$$e = (4.69 \pm 0.1) \times 10^{-10} \text{ esu}$$

where the values of e are calculated‡ from A ; they are in good agreement with the accepted value

$$e = 4.803 \times 10^{-10} \text{ esu}$$

3. The Frank-Hertz Experiment

3.1 GENERAL

From the early spectroscopic work it was clear that atoms emitted radiation at discrete frequencies; from Bohr's model the frequency of the radiation ν is related to the change in energy levels through $\Delta E = h\nu$. Further experiments demonstrated that the absorption of radiation by atomic vapors also occurred only for discrete frequencies.

It is then to be expected that transfer of energy to atomic electrons by any mechanism should always be in discrete amounts§ and related to the atomic spectrum through the equation given above. One such mechanism of energy transfer is through the inelastic scattering of electrons from the

† D. Peters, class of 1962.

‡ It is seen that in this special case (partly because of the low voltage) the diameter of the drops is so small that the correction to the Stokes equation is considerable (15 percent).

§ They are still bound after the process.

3. The Frank-Hertz Experiment

entire atom. If the atom that is bombarded does not become ionized, and since little energy is needed for momentum balance, almost the entire kinetic energy of the bombarding electron can be transferred to the atomic system.

Frank and Hertz in 1914 set out to verify these considerations—namely, that (a) it is possible to excite atoms by low-energy electron bombardment, (b) that the energy transferred from the electrons to the atoms always had discrete values, and (c) that the values so obtained for the energy levels were in agreement with the spectroscopic results.

The necessary apparatus consists of an electron-emitting filament and an adequate structure for accelerating these electrons to a desired (variable) potential. The accelerated electrons are allowed to bombard the atomic vapor under investigation and the excitation of the atoms is studied as a function of accelerating potential.

For detecting the excitation of the atoms in the vapor it is possible to observe, for example, the radiation emitted when the atoms return to the ground state, or the change in absorption of a given spectral line, or some other related phenomenon; however, a much more sensitive technique consists in observing the electron beam itself. Indeed, if the electrons have been accelerated to a potential just *equal* to the energy of the first excited level, some of them will excite atoms of the vapor and as a consequence will lose almost all their energy; clearly, if a small retarding potential exists before the collector region, electrons that have scattered inelastically will be unable to overcome it and thus will not reach the anode.

These conditions are created in the experimental arrangement by using two grids between the cathode and collector. When the potentials are distributed as in Fig. 1.5a, the beam is accelerated between the cathode and grid 1; then it is allowed to drift in the interaction region between the two grids and has to overcome the retarding potential between grid 2 and the anode. When the threshold for exciting the first level is reached, a sharp decrease in electron current is observed, proportional to the number of collisions that have occurred (product of atomic-density and cross section). It is clear that when the threshold of the next level is reached, a further dip in the collector current will be observed. These current decreases (dips) are superimposed on a monotonically rising curve; indeed the number of electrons reaching the anode depends on V_{acc} , inasmuch as it reduces space charge effects and elastic scattering in the dense vapor. In addition, the dips are not perfectly sharp because of the distribution of velocities of the thermionically emitted electrons, and the rise of the excitation cross section.

An alternate distribution of potentials is shown in Fig. 1.5b, where V_{acc} is applied at grid 2 so that an electron can gain further energy after a col-

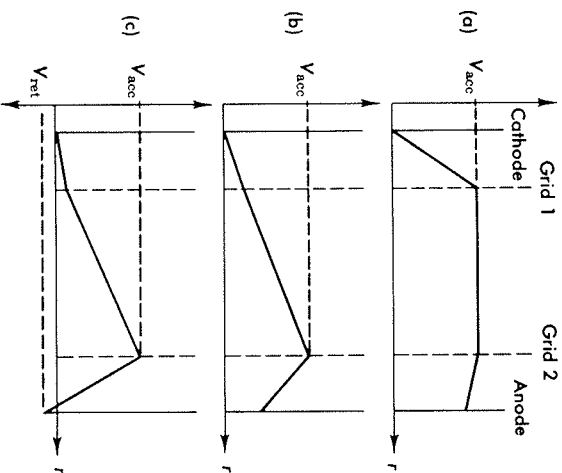


FIG. 1.5 Different configurations of the potential in a Frank-Hertz arrangement. (a) For observation of single excitation. (b) For observation of multiple excitation. (c) For measuring the ionization potential.

lision in the space between the two grids. In this case when V_{acc} reaches the first excitation potential, inelastic collisions are again possible and the decrease in electron current is observed at the anode; when, however, V_{acc} reaches a value twice that of the first excitation potential, it is possible for an electron to excite an atom halfway between the grids, lose all its energy, and then gain anew enough energy to excite a second atom and end with practically zero energy at grid 2; thus it is not able to overcome the retarding potential and reach the anode, giving rise to a second dip in the current.

The advantage of this setup is that the current dips are much more pronounced, and it is easy to obtain fivefold or even larger multiplicity in the excitation of the first level. However, it is practically impossible to observe the excitation of higher levels. As before, a slight retarding potential is applied between grid 2 and the anode, and an accelerating potential between the cathode and grid 1, sufficient to overcome space charge effects and to provide adequate electron current. It is evident that the density of the atomic vapor through which the electron beam passes greatly affects the observed results. Low densities result in large electron currents but very small dips; on the contrary, high density has as a consequence weaker currents but proportionally larger dips. When mercury vapor is used, obviously the adjustment of the tube temperature provides control of the density.

Another important point is that in principle the experiment must be

performed with a monatomic gas; since if a molecular vapor is bombarded, it is possible for the electrons to transfer energy to the molecular energy levels which form almost a continuum. Some of the preferred elements for the Frank-Hertz experiment are mercury, neon, and argon.

The same apparatus can be used for the measurement of the ionization potential—that is, the energy required to remove an electron completely from the atom. In this case, instead of observing the bombarding electron beam, it is easier to detect the ions that are formed. The distribution of potentials is as shown in Fig. 1.5c, where the anode is made slightly negative with respect to the cathode; no electrons can then reach the anode, which becomes an ion collector. The accelerating potential is increased until a sharp rise in the ion current measured at the anode is observed.

In both types of measurements the values obtained for the accelerating potential have to be corrected for the contact potential difference (cpd) between cathode and anode.[†] If in the excitation experiment the same level has been observed two or more times, however, the potential difference between adjacent potential difference shifts the whole voltage scale. Once the excitation energy has been found the contact potential difference is given by the difference between this true value and the first peak; in turn the contact potential difference so found can be used to correct the ionization potential measurement.

3.2 THE EXPERIMENT

In this laboratory a mercury-filled tube made by the Leybold Company (55580) is used; the electrode configuration is shown in Fig. 1.6; the circuit diagrams for the measurement of excitation and of ionization potential are given in Figs. 1.7a and 1.7b respectively.

As can be seen from the circuit diagram, grid 1 is operated in the neighborhood of 1.5 V, and the retarding potential is of the same order. The anode currents are of the order of 10^{-9} amp and are measured either with a sensitive galvanometer (for example Leeds and Northrup No. 2500) or with a Keithley 600A electrometer (see Chapter 4); adequate shielding of the leads is required to eliminate a-c pickup and induced voltages. The diagram of Fig. 1.7a uses the distribution of potentials as shown in Fig. 1.5b and the accelerating voltage can be measured with an ordinary voltmeter (for example, Triplet 625) in steps of 0.1 V, or with a vacuum tube voltmeter.

[†] See Chapter 3. Briefly this is because the “work function” for the metal of which the anode is made is usually higher than that of the cathode. The work function is a measure of the “ionization potential” of the metal; that is, of the energy needed to extract an electron from it.

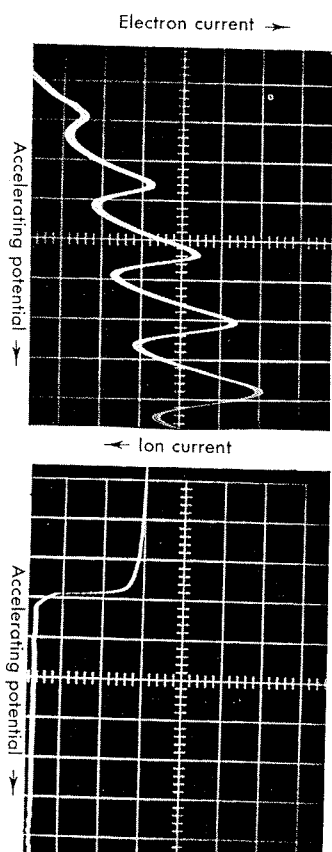


Fig. 1.9 Oscilloscope display of Frank-Hertz experiment. (a) Beam current vs. accelerating potential. (b) Ion current vs. accelerating potential.

the tube is too cool, the emission current will be large, and the maxima, particularly those of higher order, will be washed out.

With present-day techniques it is possible to use an oscilloscope for a simultaneous display of the electron or ion current against accelerating potential. In this laboratory a Tektronix 545 oscilloscope was used; its sweep generator (sawtooth) output is fed to the accelerating grid, while it synchronously drives the horizontal sweep; the output of the Keithley is fed to the vertical input. An excitation curve as well as an ionization curve obtained by a student† in this fashion are shown in Fig. 1.9. The oscilloscope method can be very useful in finding optimum operating conditions for mercury vapor pressure and electron beam density.

3.3 ANALYSIS OF THE DATA

Two sets of data obtained by a student‡ for the excitation potential point by point are shown in Fig. 1.10; both curves are obtained at a temperature of 195°C and with +1 V on grid 1. The filament voltage is 2.5 V for curve C and 1.85 V for curve D with the consequent decrease of the electron current by a whole decade.

Readings are taken for 1-V changes on grid 2 with smaller steps in the vicinity of the peak. A significant decrease in electron (collector) current is noticed every time the potential on grid 2 is increased by approximately 5 V, thereby indicating that energy is transferred from the beam in (bundles) "quanta" of 5 eV only. Indeed, a prominent line in the spectrum of mercury exists at 2537 Å, corresponding to $12378/2537 = 4.86$ eV, arising from the

† D. Statt, class of 1963.

‡ D. Owen, class of 1963.

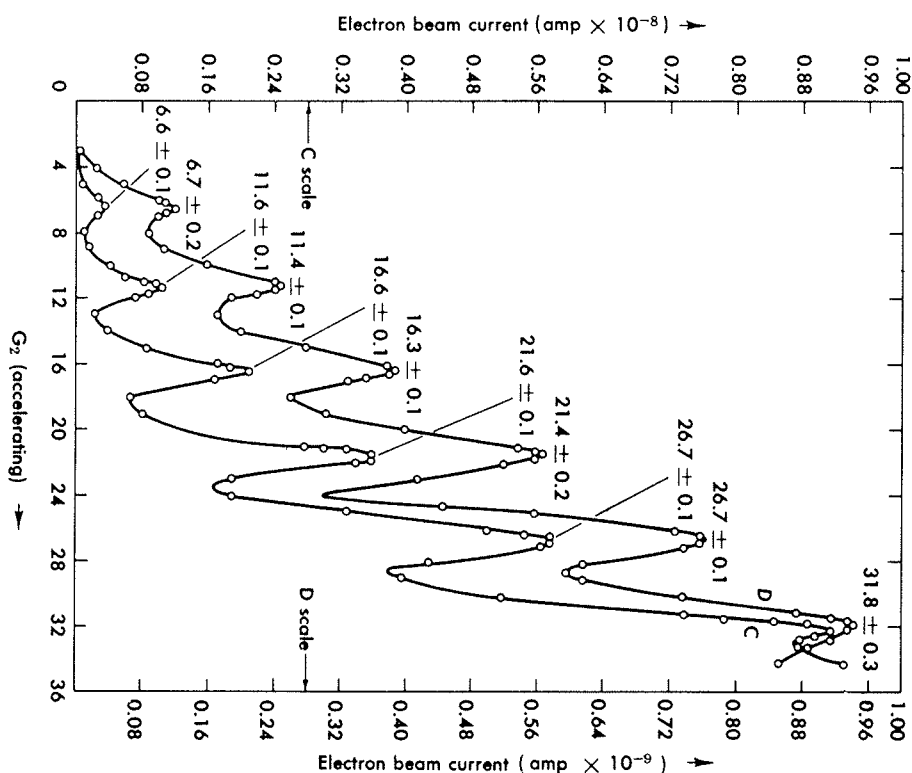


Fig. 1.10 Plot of beam current versus accelerating voltage in Frank-Hertz experiment. Curve C (left-hand scale) is obtained with the filament set at 2.5 volts while Curve D (right-hand scale) with filament at 1.85 volts.

transition of the $6s6p\ ^3P_1$ excited state to the $6s6s\ ^1S_0$ ground state.† Thus our interpretation is that the electrons in the beam excite the mercury atom from the ground state to the 3P_1 state, thereby losing 4.86 eV in the process.

The location of the peaks is indicated in Fig. 1.10 and was measured in this case with a vacuum tube voltmeter (VTVM). The average value ob-

† See Chapter 2.

tained for the spacing between peaks is

$$5.02 \pm 0.1 \text{ V}$$

to be compared with the accepted value of

$$4.90 \text{ V}$$

while the spectroscopic value for the energy level difference (as mentioned before) is 4.86 eV.

Using the value found for the spacing between peaks and the location of the first peak, we obtain the contact potential:

$$(6.65 \pm 0.15) - (5.02 \pm 0.1) = 1.63 \pm 0.18 \text{ V}$$

As mentioned in Section 3.1, with the configuration of potentials used (Fig. 1.5b) it is more probable that the same energy level will be excited twice rather than that several different levels will be excited; indeed this is the way in which the data of Fig. 1.10 have been interpreted. This is not surprising if one considers the excitation probabilities for the energy levels lying closest to the ground state of mercury. It is possible, however, by using different grid and voltage configurations (for example, Fig. 1.5a) and improved resolution, to observe the excitations to other levels, namely, the 6^1P_2 , 6^3P_0 and 6^1P_1 .

For the ionization potential, data obtained by a student† are shown in Fig. 1.11. A word of caution is to be added to the interpretation of such ionization curves, which seem strongly dependent on filament voltage and vapor pressure: indeed the very sharp increase observed in ion current is due to an avalanche (regenerative effect) of the ejected electrons ionizing more atoms, the thus-ejected electrons ionizing still more atoms and so on; this avalanche does not necessarily occur as soon as the ionization threshold is crossed. If the vapor is too dense, the ions recombine before reaching the anode, thus masking the effect until complete breakdown sets in.

The curve shown was taken at a temperature of 155°C with a filament voltage of 2.6 V. If, then, the onset of ion current is taken to be at $11.4 \pm 0.2 \text{ V}$, and using the value for the contact potential previously determined (from the excitation curve), $1.63 \pm .18 \text{ V}$, the ionization potential is obtained as

$$(11.4 \pm 0.2) - (1.63 \pm 0.18) = 9.77 \pm 0.25 \text{ eV}$$

only in fair agreement with the accepted value of 10.39 eV.

An additional feature of the curve of Fig. 1.11 is a "knee" in the ion current, setting in at approximately 8 V; the observation of this "knee" as well is strongly dependent on the temperature and current density, but

† J. Reed, class of 1961.

can be consistently reproduced over a considerable range of these parameters. In order to understand this behavior we remember that the arrival of ions at the anode is equivalent to the departure of electrons; indeed the observed behavior is due to a photoelectric effect produced at the anode, by short-wavelength light quanta (the electrons are further accelerated by grid 2). When the electron beam reaches 8 V, it can excite the 6^1P_1 level (lying at 6.7 eV above the ground state, plus 1.63 V for contact potential difference), so the mercury atoms radiate the 1849 Å ultraviolet line when returning to the ground state. These quanta are very efficient in ejecting photoelectrons, and the cylindrical geometry of the anode is most favorable for this process.

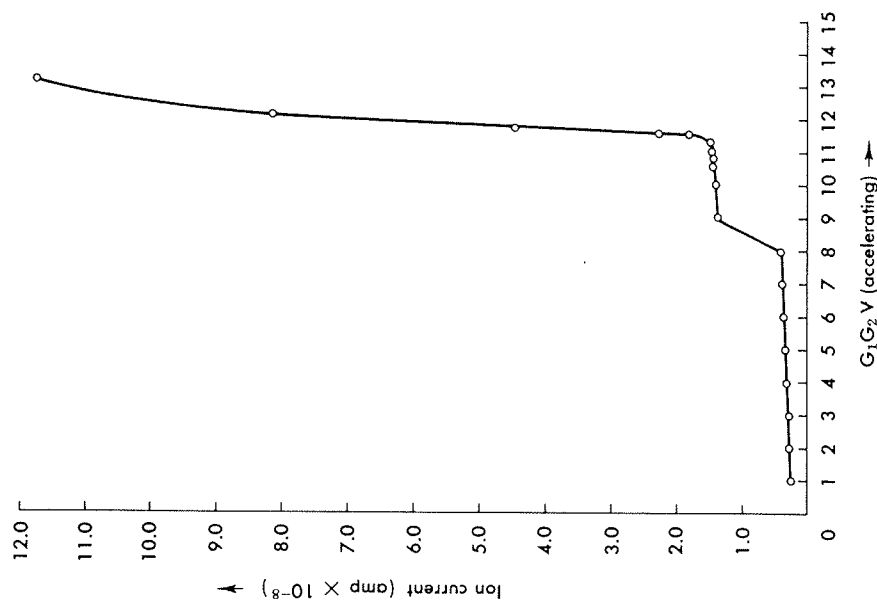


Fig. 1.11 Ion current versus accelerating voltage in Frank-Hertz experiment. Knee at 8 V is due to photoeffect.

4. The Photoelectric Effect

4.1 GENERAL

It was observed as early as 75 years ago that most metals under the influence of radiation (light), especially ultraviolet radiation, emit electrons. This phenomenon was termed photoelectric emission, and detailed study of it has shown:

(a) That the emission process depends strongly on the frequency of the light, and that for each metal there exists a critical frequency such that light of lower frequency is absolutely unable to liberate electrons while light of higher frequency always does. Indeed, for a given surface, if the frequency of the incident radiation is increased, the energy of the emitted electrons increases in some linear relation.

(b) The emission of the electrons occurs within a very short time interval after the arrival of the radiation, and the number of electrons emitted is strictly proportional to the intensity of the radiation.

The experimental facts given above are among the strongest evidence for our present-day belief that the electromagnetic field is *quantized*. They cannot be explained in terms of a continuous energy distribution in the radiation field, but it must be assumed that the field consists of "quanta" of energy

$$E = h\nu$$

where ν is the frequency of the radiation and h is Planck's constant (an expression we have already used in Section 3). These quanta are called *photons*.

Further it is assumed† that the electrons are bound inside the metal surface with an energy $e\phi$, where ϕ is called† the "work function," and that all such electrons have equal probability of absorbing a photon. It then follows that if the frequency of the light ν is such that

$$h\nu > e\phi$$

it will be possible to eject photoelectrons, while if

$$h\nu < e\phi$$

this is impossible, since the probability that an electron will absorb two photons simultaneously is minimal. In the former case, the excess energy of the quantum appears as kinetic energy of the electron, so that

$$h\nu = \frac{1}{2}mv^2 + e\phi \quad (4.1)$$

† See Chapter 3.

which is the famous photoelectric equation formulated by Einstein in 1905. In writing the equation in this form we express the fact that the energy is shared between the electron and the photon only; however, to balance momentum† a third body is needed, which in this case is the crystal lattice, which recoils with negligible energy.

Thus in the photoelectric effect we observe the transfer of the total energy of a photon to an electron bound in a lattice. We will see later another transfer mechanism, prevalent at higher photon energies, whereby only part of the energy of the photon is transferred to a free electron: this is the Compton scattering.

Equation 4.1 has been extensively verified for many materials and over a broad range of frequencies. What is experimentally measured is the energy of the emitted photoelectrons against frequency, either with a magnetic field or in a simpler way by a retarding potential technique, as is done in this laboratory exercise. Since the "work function" ϕ is usually not known beforehand, the kinetic energy of the photoelectrons $E_e = \frac{1}{2}mv^2$ is obtained as a function of ν so that the slope of the straight line

$$E_e = h\nu - e\phi$$

yields h , and the intercept at the extrapolated point $\nu = 0$, can give $e\phi$. When a retarding potential V is used to measure E_e , we have $E_e = eV_0$, so that really it is the ratio h/e that is determined:

$$V_0 = (h/e)\nu - \phi \quad (4.2)$$

The arrangement generally used consists of a clean surface of the metal to be investigated, and an anode facing or surrounding the cathode, both sealed in vacuum. When radiation is incident on the cathode, electrons are emitted which reach the anode giving rise to a detectable current if the circuit between anode and cathode is completed through a sensitive current meter as shown in Fig. 1.12. If a negative potential V is applied to the anode, only electrons with $E_e > eV$ can reach the anode, and for some potential V_0 no electrons at all arrive at the anode; this retarding potential multiplied by e is equal to the energy of the fastest electrons emitted. In practice all electrons are not emitted with the same energy, and therefore the threshold at V_0 is not very sharp; space charge effects further reduce the definition.

An additional consideration, already encountered in the Frank-Hertz experiment is the contact potential difference; namely, the fact that the potential applied and measured across the anode and cathode leads does

† Note that in any event the electrons are emitted in a direction opposite to that of the incoming radiation.

not equal the potential that the electron traveling from the cathode to the anode has to overcome. To see this, consider Fig. 1.13, where ϕ_c represents the work function of the cathode and $\phi_A \neq \phi_c$ is the work function of the anode. The external voltage V' is applied between metallic junctions and we may neglect the ohmic voltage drop in the leads; thus the electrons *inside* the anode are at potential V' higher than the electrons *inside* the cathode. The energy losses around the loop of Fig. 1.13 must, however, be zero, and the arrows indicate the direction for which an electron *loses* energy in the field (namely, negative potential); if V is the potential seen by the free electron, we obtain

$$-\phi_c + eV + e\phi_A - eV' = 0$$

or

$$V = V' - (\phi_A - \phi_c) \quad (4.3)$$

The term $(\phi_A - \phi_c)$ is the contact potential difference (cpd) and usually $\phi_A > \phi_c$. Therefore the measured potentials V' must be corrected according to Eq. 4.3 in order to be used in Eq. 4.2. One way of finding the contact potential difference is to normalize all curves to the same saturation current and observe for what (common) value of V' saturation sets in; this must correspond to the point where V changes over from retarding to accelerating—namely, from Eq. 4.3, $V = 0$ or $V' = \phi_A - \phi_c = \text{cpd}$.

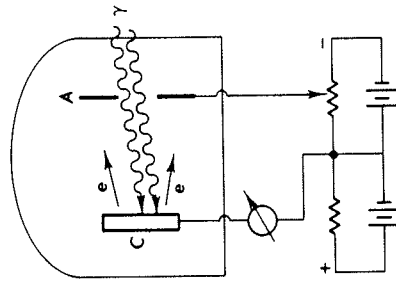


Fig. 1.12 (Left) Schematic of a setup for detecting the photoelectric effect; the anode can be made either negative or positive with respect to the cathode.

Fig. 1.13 (Right) Potential at anode (—) and cathode (+) of photoelectric cell; w and w' are the work functions of anode and cathode, respectively. Note that $\phi - V - \phi' + eV' = 0$ so that the potential seen by the free electron is $V = V' - (\phi - \phi')$.

By combining Eqs. 4.1 and 4.3 we note that if emission stops for an applied retarding potential V_0' , then

$$V_0 = V_0' + (\phi_A - \phi_c) \quad (4.3b)$$

further

$$e|V_0| = \frac{1}{2}mv^2 = h\nu - e\phi_c \quad (4.1b)$$

so that

$$|V_0'| = (h/e)\nu - \phi_A \quad (4.2b)$$

namely, a plot of the *applied* stopping potential (without cpd corrections) vs. ν yields a line of slope h/e but with an intercept at $\nu = 0$ equal to the work function of the *anode* rather than to the work function of the cathode predicted by Eq. 4.2.

4.2 THE EXPERIMENT

To perform the experiment we need a source of monochromatic light, at several frequencies, the photoelectric cell, and a sensitive current detecting device.

In this laboratory the photocell is of special construction (Leybold catalogue 55877) with a potassium (coated) cathode and an anode which consists of a platinum ring (Fig. 1.14). A special casing is available for pro-

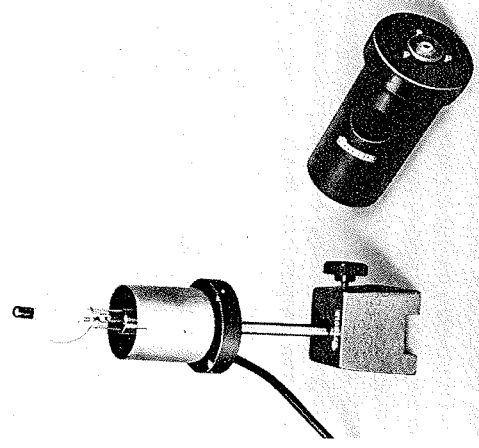


Fig. 1.14 A photocell made by the Leybold Company (Catalog No. 55877).

tection, electrostatic shielding, and adequate insulation of the anode contact.

The optical system is mounted on an optical bench as shown in Fig. 1.15. The light source is a high-pressure mercury discharge (Will Corp. 17391); it is focused onto slit 1 and then dispersed in the direct vision prism (Leybold 46604). Slit 2 selects the desired wavelength. The direct vision prism consists of a combination of two crown glass and one flint glass prisms and passes medium wavelengths without refraction.

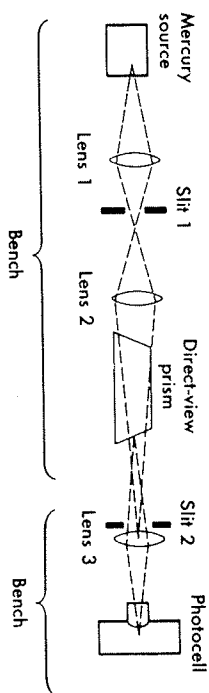


Fig. 1.15 Optical system for photoelectric effect measurement.

When the system is set up, the source must be focused on slit 1, and the image of the slit focused on lens 3; a sharp image of the spectral line can then be obtained on the photocathode by adjusting the prism and photocell positions. Since the system is chromatic, each line must be refocused independently. Obviously it is easier to make these adjustments in a darkened room or with the help of black cloth.

Care must be taken that the incident radiation contains only the line chosen for the investigation, and that it does not hit the platinum anode.[†] This can be achieved by placing an appropriate mask in front of the photocell.

The mercury lines most readily available are

Yellow	at	5770 Å and 5790 Å
Green	at	5461 Å
Blue-green	at	4916 Å (weak)
Blue	at	4358 Å mainly, and at 4343, 4339 Å
Violet	at	4047 Å

The electrical connections are shown in Fig. 1.16. The photocurrent is measured with a Keithley electrometer connected to the photocathode while the anode is returned to ground through a variable d-c voltage which provides the desired retarding or accelerating field. Since the voltages applied between anode and cathode are very small, appropriate corrections must be made for the voltage drop produced in the meter, due to the photocurrent flowing in the large input impedance.

[†] Due to scattering a small fraction of the incident radiation reaches the anode.

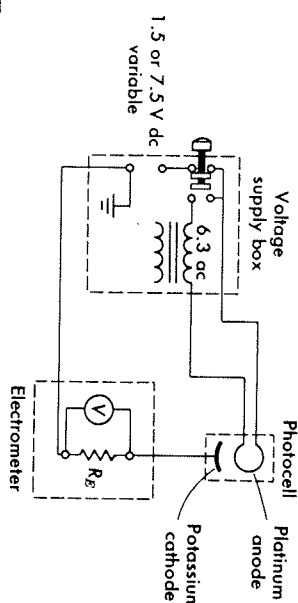


Fig. 1.16 Wiring diagram for photoelectric effect measurement.

Provisions exist for connecting the anode ring to a 6.3-V a-c supply; thus the anode may be heated in order to evaporate any potassium traces that have deposited on it. It is usually adequate to close the circuit momentarily, since long heating will destroy the anode. The retarding (or accelerating) potential is obtained from a fixed source and a calibrated resistive network, which must, however, be checked against a potentiometer.

In performing the experiment, after the desired line has been focused on the photocathode, the current is measured as a function of the voltage applied between anode and cathode. In principle the accelerating potential should be increased (in appropriate steps) until saturation is reached; this is, however, difficult to achieve with the photocell described here.[†] The decelerating potential should also be decreased until zero current is observed and beyond that point, to assure that the current remains zero.

Frequently, as is the case with this particular apparatus, a reverse current is observed leveling off at approximately 10^{-12} amp; this is attributed to photoemission from the anode and introduces difficulties in the exact determination of the stopping potential V_0' . Nevertheless, it is possible to obtain significant results if the same consistent criterion is used at all wavelengths for obtaining V_0' .

Since the current variations in the vicinity of the stopping potential are of the order of 10^{-12} amp, their measurement becomes difficult and special care must be exercised. Leakage current across the glass face of the tube must be minimized, as by surrounding the bulb base with a moisture absorber (silica gel), especially on humid days. The cathode contact must be kept very clean (with alcohol and hot air), all leads on the high-impedance side must be coaxially shielded, and the appropriate connectors must be used. The electrometer and apparatus should be protected from vibrations and stray field pickup. Dark current must be monitored and appropriate corrections applied.

[†] Mainly due to geometrical considerations.

Finally, caution must be exercised in using the mercury source, since its envelope transmits ultraviolet light, which can cause serious damage to the eyes and sunburn to the skin.

4.3 ANALYSIS OF THE DATA

The data presented below were obtained by students.[†] The five lines of mercury mentioned in the previous section were used, and the photocurrents near saturation due to these lines were in the following proportion:

Yellow	1.00
Green	1.50
Blue-green	0.44
Blue	1.70
Violet	0.55

These yields are a combination of the intensity of the spectral lines, their attenuation in the optical system, and the photosensitivity of the cathode, which is not the same at all wavelengths. In analyzing the data as mentioned before it is useful to normalize all photocurrents to the same saturation

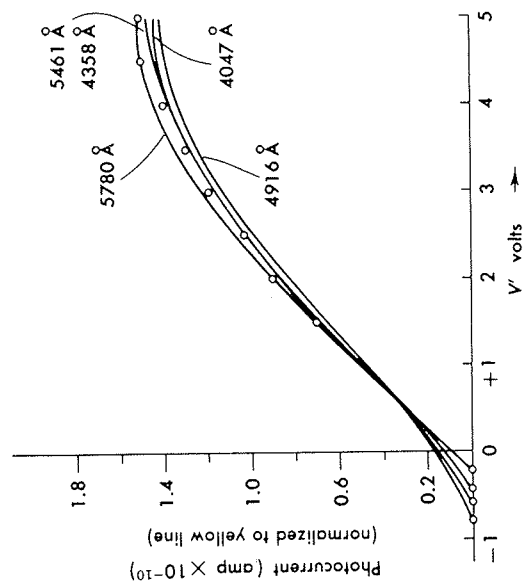


Fig. 1.17 Photocurrent as a function of anode voltage. The currents have been normalized to the yield from the yellow lines ($\lambda = 5780 \text{ \AA}$).

[†] D. Owen and D. Sawyer, class of 1963.

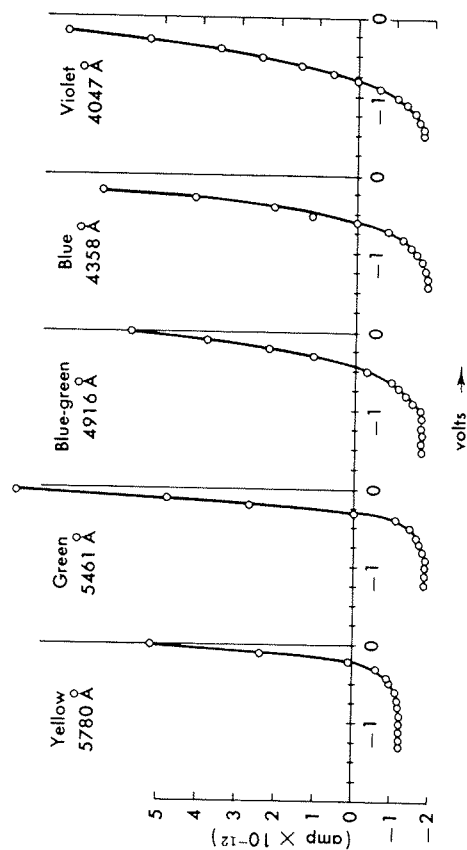


Fig. 1.18 Photocurrent for different wavelengths as a function of stopping potential; the same normalization as in Fig. 1.17 has been used. Note the reverse current due to photoemission from the anode.

tion value; the results of such normalization for the five wavelengths are shown in Fig. 1.17. We note that in the accelerating region the curves are quite similar, and the small differences can be attributed to observational errors.

On the other hand, Fig. 1.18 represents the region close to the stopping point, but separately for each wavelength; the normalized photocurrents are shown. In spite of the reverse current it is possible to read off the stopping potential for each line; the difficulty arises rather from the apparent zero slope of the curves.[†] From these curves values of V_0' have been obtained (a) by forming the intersection of the tangents to the limiting branches of the curves, and (b) by estimating the voltage at which the current curve begins to rise. These values are given in Table 1.2.

A plot of these stopping voltages and the least-squares fit are shown in Fig. 1.19. We see that in both cases a correct order of magnitude of h/e is obtained namely:

$$\text{Method (a)} \quad h/e = (3.84 \pm 0.55) \times 10^{-15} \text{ V-sec} \\ \text{intercept at } \nu = 0 \quad V' = +1.2 \text{ V}$$

$$\text{Method (b)} \quad h/e = (3.84 \pm 0.4) \times 10^{-15} \text{ V-sec} \\ \text{intercept at } \nu = 0 \quad V' = 1.6 \text{ V}$$

[†] It should be a finite slope, but this is not observed in the present arrangement because of the reverse current.

TABLE 1.2
RETARDING POTENTIALS REQUIRED TO STOP PHOTOEMISSION AS OBTAINED FROM
FIG. 1.18

Line	Retarding potentials	
	(a)	(b)
Yellow	-0.25 V	-0.7 V
Green	-0.40	-0.8
Blue-green	-0.70	-1.0
Blue	-0.82	-1.3
Violet	-1.15	-1.5

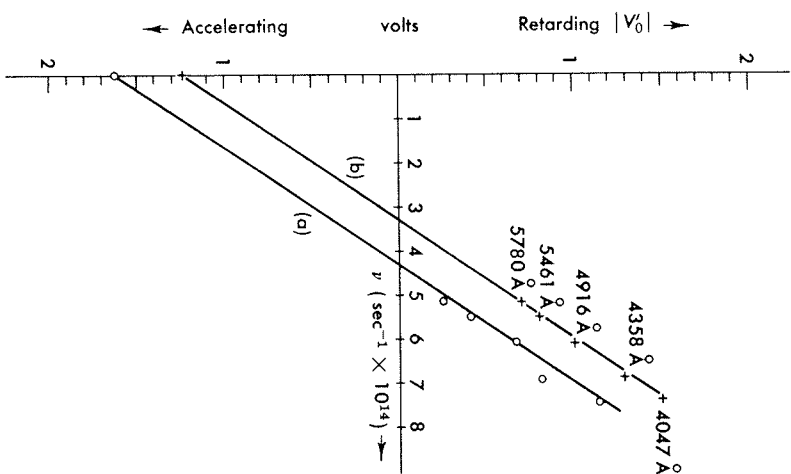


FIG. 1.19 Plot of applied stopping potential versus frequency. Lines (a) and (b) refer to the data of Table 1.2.

4. The Photoelectric Effect

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Multiplying by the charge of the electron $e = 1.6 \times 10^{-19}$ coulombs we obtain

$$h = (6.14 \pm 0.8) \times 10^{-34} \text{ joules-sec}$$

to be compared with the accepted value of

$$h = 6.61 \times 10^{-34} \text{ joules-sec}$$

While the value obtained for h/e is quite satisfactory it is not possible to draw any conclusions with regards to the cpd. The known values for the work functions are

$$\text{anode, platinum } \phi_A = 5.29 \text{ V}$$

$$\text{cathode, potassium } \phi_C = 2.15 \text{ V}$$

And thus $\text{cpd} = 3.14 \text{ V}$. This value is in qualitative agreement with the saturation data of Fig. 1.17. However, from Fig. 1.18 we are inclined to deduce $\phi_A \approx 1.5 \text{ V}$ rather than $\approx 5 \text{ V}$; this fact, is a further indication that cathode material had deposited on the anode while these data were obtained.

Thus we have seen in three basic experiments that fundamental quantities of nature, such as the electric charge, the energy of electrons bound in an atom, and the energy of the electromagnetic field are quantized: that is, they cannot take any of a continuous set of values but only discrete ones. This fundamental characteristic of our world was first formulated in 1901, when Max Planck introduced it as the basic hypothesis for his theoretical interpretation of the spectrum (continuous in frequency) of a heated black body. It has led to a serious revision of both the method of thought and the mathematical tools of physics.

A slightly more detailed description of these experiments, including references to the original literature can be found in G. P. Harwell and J. J. Livingood, *Experimental Atomic Physics*. New York: McGraw-Hill (1961).